INNOPOLIS UNIVERSITY

SNE MASTER TRACK

COURSE: [F25] SECURITY OF SYSTEMS AND NETWORKS

SSN Lab 1 Assignment

Classical Crypto

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Cryptography: In the modern sense, cryptography is a science and technology that uses mathematical algorithms and keys to make information secret, unbreakable, verifiable, and non-repudiable [1][2].

• **Plaintext:** A secret message is converted into ciphertext using a special algorithm and key.

• The **ciphertext** is then converted back into the original message, plaintext, using the key.

Purpose of Cryptography: The main purpose of cryptography is based on four pillars:

1. Confidentiality: Only authorized persons can read the message. For example, customer information is encrypted in a bank's online transfer system [1].

2. Integrity: The main goal of this pillar is to detect whether any changes have been made to the message being sent. For example, checksums are generated using hash functions in e-mails or digital documents [3].

3. Authentication: This stage verifies whether the message actually came from the sender. For example, proof is obtained through the use of digital signatures [4].

4. **Non-repudiation:** In this, the sender of the message cannot later deny that he did not send the message. This is important in e-commerce or legal contracts [5].

Brief history:

- **World War II:** The German Enigma Machine and the role of British cryptographers including Alan Turing in breaking it paved the way for modern cryptography [6].
- **Modern era:** Developments in computer science led to the development of algorithms such as RSA, AES, Diffie-Hellman, and what is now the mainstream of quantum cryptography research [7].

Asymmetric Cryptography: Asymmetric Cryptography or Public-Key Cryptography is a cryptographic method that uses two separate but mathematically related keys [8]:

- **Public Key:** It is publicly shared and anyone can encrypt with it.
- **Private Key:** It is secret only to the owner and can be easily decrypted with it.

Mathematically, if P is plaintext and C is ciphertext, then:

$C = E_{Kpublic}(P)$, and $P = D_{Kprivate}(C)$

Here, a message encrypted with a public key can only be decrypted with the corresponding private key. This dual-key structure solves the key distribution problem inherent in symmetric cryptography [9].

History and Development:

• **Diffie–Hellman:** Whitfield Diffie and Martin Hellman first published the public-key concept in 1976. This is considered the beginning of asymmetric cryptography [9].

• **RSA:** Rivest, Shamir, and Adleman developed the RSA algorithm in 1978, which is still considered one of the leading public-key systems today [10].

• **Elliptic Curve Cryptography (ECC):** Victor Miller and Neal Koblitz independently proposed ECC in 1985, which is capable of ensuring security for small keys [11].

Mathematical basis: Asymmetric cryptography generally relies on number theory and the inability to solve complex mathematical problems.

The security of RSA depends on the complexity of factoring large numbers $n = p \times q$ into prime numbers [3]. The security of ECC depends on the complexity of the Elliptic Curve Discrete Logarithm Problem [11].

Advantages:

- No risk of key distribution (Public Key can be kept public).
- Can be used for digital signatures and authentication.
- Ensures non-repudiation.

Limitations:

- Slower than symmetric cryptography.

- This type of cryptography is ineffective for directly encrypting large data.
- High computational cost.

Practical applications:

- **SSL/TLS Certificates:** Used to secure web browsing.
- **PGP/GPG:** Used for secure email communication.
- **Cryptocurrency:** Used in Bitcoin and other blockchains.
- **Digital signatures:** Used in e-governance, software distribution, e-commerce contracts, etc.

Cipher: The basic foundation of cryptography is the cipher. The word "cipher" comes from the word "ṣifr", which means "empty" or "secret". A cipher is a type of algorithm that converts a readable message (plaintext) into an unknown or unreadable message (ciphertext) using a special rule [12].

Mathematically, a cipher consists of two basic functions:

- **Encryption Function (E):** $C = E_K(P)$
- **Decryption Function (D):** $P = D_K(C)$

Here,

P = Plaintext (original message)

C = Ciphertext (secret message)

K = Key

Types of Ciphers: Ciphers can generally be divided into two large groups-

1. **Classical Cipher**
2. **Modern Cipher**

Classical Cipher: The various cipher systems used in ancient times are called Classical Ciphers. They are relatively simple and were usually done by hand or using pen and paper.

a) Substitution Cipher

Here, each letter or symbol is replaced by another letter/symbol.

- Caesar Cipher: Each letter is moved a certain number of steps. For example, if "INFORM" is moved 4 steps, the cipher text is " MRJSVQ " [13].
- Monoalphabetic Cipher: One letter is replaced by another specific letter.
- Polyalphabetic Cipher (Vigenère Cipher): The message is encrypted using more than one alphabet.

b) Transposition Cipher

Here the position of the letters is changed, but the letters remain unchanged.

- **Rail Fence Cipher:** In this type of cipher, the message is arranged in a wavy manner and then read in a straight line.
- **Columnar Transposition:** In the columnar transposition cipher, the letters of the message are arranged in a table form and read in the order of specific columns.

Modern Cipher: With the development of computer science, modern ciphers have been created based on complex mathematical algorithms.

A) Block Cipher: Data is divided into specific blocks and each block is encrypted with a key. Examples: DES, AES [14].

B) Stream Cipher: Data is encrypted bit by bit or byte by bit. Example: RC4 [15].

C) Asymmetric Cipher: Encryption and decryption are performed using public and private keys. Examples: RSA, ECC [16].

Practical examples of ciphers

- Banking sector: AES is used to protect customer passwords and transaction data.
- Email encryption: RSA is used in PGP (Pretty Good Privacy).
- Military communications: Vigenère Cipher was used in the past, but more advanced algorithms are used today.
- Web security: HTTPS protocol uses a combination of RSA and AES.

Importance of Cipher: This method protects the confidentiality of information, so that sensitive messages or data cannot be read or modified by unauthorized outsiders. In addition, if the information is altered in any way, it can be easily detected, which ensures the integrity of the information. Using this method in digital transactions ensures the credibility of the transaction and reduces bias or fraud. In addition, it plays an important role in preventing cyber-attacks, thereby protecting the system from unauthorized access or data theft.

Common Cipher Tools: In implementing cryptography, it is not enough to simply know the algorithms; it is also very important to use effective cipher tools. These tools are available as computer or online software and allow users to easily encrypt or decrypt messages, create digital signatures, and test various ciphers. They are used for educational, research, and professional security purposes [17], [18].

1. **Online Cryptography Tools**
2. **dcode.fr**

- dcode.fr is a widely used online cipher tool that supports various types of encryption/decryption.
- Provides simple demos and online solutions for classical ciphers such as Caesar, Vigenère, Substitution Cipher, as well as modern ciphers.
- Students or researchers can easily encrypt/decrypt text and analyze ciphers [19].

3. **Online Cipher Solver Tools**

- **quipqiup.com:** Can solve classical substitution or monoalphabetic ciphers quickly.

- **cryptii.fr (Cryptii online):** Supports online encryption/decryption of text-to-binary, Base64, Morse Code, etc.
- **rumkin.com/tools/cipher:** Can be used to solve classical ciphers or simple encryption.

4. **CrypTool 2.1** is an open source educational cryptography software. It is GUI-based and allows students to easily test various algorithms (DES, AES, RSA, Vigenère, Caesar). In addition, frequency analysis, block cipher, and digital signature can be tested [17], [18].

Practical Applications

In education and training, CrypTool 2.1 and dcode.fr are used to easily understand the concepts of various ciphers and cryptanalysis. In the analysis of classical ciphers, dcode.fr and -erquipqiup.com provide quick solutions, which are especially helpful for practicing. In addition, CrypTool is used to test Symmetric and Asymmetric Cryptography in data security verification, which helps users ensure the security of the system.

Enigma Machine: Germany's Secret Message Protection:

The Enigma machine was an electro-mechanical rotor machine used by the German military during World War II to encrypt messages [19], [20].

The main purpose of this machine was to convert messages from plaintext to completely unbreakable ciphertext. Messages generated by the Enigma machine would be almost impossible to decrypt outside the German military.

How the Germans used Enigma: Each message was encrypted using a Rotor and Plugboard, where each rotor rotated, changing the encryption of each letter. The message key, determined by the position of the Rotor and the Plugboard settings, changed daily. As a result, both the sender and receiver had to know the correct key in order for the message to be sent and received correctly. The system was so complex that the number of possible keys for each message was in the billions, making it impossible to easily brute-force [21].

Main parts:

- Rotors (Rotating Wheels): Repositions the letters.
- Reflector: Reflects the signal inside the rotors.
- Plugboard (Steckerbrett): Switches the pairs of letters.

Alan Turing's Bombe Machine: Alan Turing and his colleagues at Bletchley Park developed a mechanical solution to break Enigma messages.

Name of the machine: Bombe

How the Bombe Machine worked: Each message was encrypted using a Rotor and Plugboard, which helped keep the message secret. As each rotor turned, the encryption of each letter changed, so that the same letter would be converted into a different form even if sent repeatedly. In addition, the key of the message, i.e. the position of the Rotor and the setting of the Plugboard, changed every day, which made the message more secure. Both the sender and the receiver had to know the correct key, otherwise the message could not be decrypted [22], [23].

Special features of the Bombe Machine: The Bombe machine used more than one hundred rotating drums, which helped simulate every possible rotor position. It was possible to test the possible keys of multiple messages simultaneously, as a result, the decryption process was completed much faster. In addition, the machine worked using electro-mechanical logic, which combined mechanical and electronic mechanisms to verify possible combinations [22], [23].

Technical aspects of the Bombe machine:

- 3–4 rotor drums were used (later expanded).
- Relays were used to test plugboard settings.
- Each drum simulated the position of a letter.
- Multiple possible keys for messages could be tested simultaneously.
- The Bombe was considered an early computer of its time [24].

Historical importance of the Bombe machine:

- During World War II, the Allied forces were able to quickly decrypt German messages.

- Turing's machine made it possible to make many military and naval decisions.

The Bombe machine used more than a hundred rotating drums, which helped simulate every possible rotor position. Multiple possible keys for messages could be tested simultaneously, making the decryption process much faster. In addition, the machine worked using electro-mechanical logic, which combined mechanical and electronic methods to verify possible combinations.

During the World War II:

During World War II, the German army sent a large number of important messages every day. Each message was encrypted by the Universal Enigma machine, which protected the German army from external information. The Allied forces realized that the strategic success of the war depended almost entirely on breaking and reading the Enigma messages [25], [26].

However, breaking the messages was not easy because:

- The key for each message changed daily.
- The rotor and plugboard provided billions of possible combinations.
- The encryption of each message changed with each letter.

That is why Alan Turing and his colleagues developed scientific methods and mechanical solutions.

Features and workings of the Bombe machine: Each rotating drum in the Bombe machine simulated the position of a rotor, which helped to verify possible combinations. Possible keys for multiple messages could be tested simultaneously, making the decryption process much faster. The machine used electro-mechanical relays, which implemented logical deduction. Automation of brute-force methods made it possible to automatically verify possible combinations without manually testing the keys. Each Bombe machine used more than 200 relays and many drums. For this reason, the Bombe was considered an early computer of its time. Using the Bombe machine, the Allied forces were able to find several new keys per day, which gave them a strategic advantage during the war. The Allied forces were able to quickly decrypt German messages with the help of

the Bombe machine, which influenced the strategic decisions of the war. The outcome of naval battles in the Atlantic and Mediterranean theaters was also changed because of this. Turing's idea and design of the Bombe machine are considered a major milestone in modern computer science and cryptography. [30] Alan Turing's Bombe machine was able to decrypt important messages during WWII, which profoundly influenced the outcome of the war and the future history of computers and cryptography [27], [28], [29].

Task 1 (a):

My teammate was **Danny C**, where I had to use the **vigenere cipher** to decode the text **ALFR BPPNHC TM F HMYPJSGPHY RCXIWD BPPNHC QIW YFP ZZYSCY BMGNB NX LZHATJLNNQC LHI MYD ZFXRPL SFLZMJHMYX WFLRYX WCLX FSB HLNYC DJJJB**

EBJ RCXIWD BPPNHC TM KFQEYW BGEB SNAP MFHJTHL FLO YAJPJ WJQJ TM STU TH SFLZMHFJP LFSEPM , where I used the key **LUFFY**

The original text was:

PRAM DEVICE IS A CONVENIENT MEMORY DEVICE FOR THE FUTURE WHICH IS NONVOLATILE AND HAS FASTER NANOSECOND RANGES READ AND WRITE SPEED

THE MEMORY DEVICE IS FASTER WITH NICE SACLING AND EVERY CELL IS NOW IN NANOSCALE RANGES

The decrypting text from my teammate:

PRAM DEVICE IS A CONVENIENT MEMORY DEVICE FOR THE FUTURE WHICH IS NONVOLATILE AND HAS FASTER NANOSECOND RANGES READ AND WRITE SPEED

THE MEMORY DEVICE IS FASTER WITH NICE SACLING AND EVERY CELL IS NOW IN NANOSCALE

Task 1 (b):

Text decryption From My Teammate:

PRAM DEVICE IS A CONVENIENT MEMORY DEVICE FOR THE FUTURE WHICH IS NONVOLATILE AND HAS FASTER NANOSECOND RANGES READ AND WRITE SPEED THE MEMORY DEVICE IS FASTER WITH NICE SACLING AND EVERY CELL IS NOW IN NANOSCALE RANGES

Texts Decryption from Teammate:

Cipher Text 1: ECIB AIPBCLU MGF **Key 1:** YOUYOUYOUYOU **Plain Text From Me:** GOOD MORNING SIR (Fig.1)

Cipher Text 2: WSN FVG RDY UTMPZ **Key 2:** PERFECT **Plain Text From Me:** HOW ARE YOU DOING (Fig.2)

Cipher Text 3: OVRGURT QWB SCS IOS **Key 3:** OKAY **Plain Text From Me:** ALRIGHT SIR SEE YOU (Fig.3)

Cipher Text 4: LK PLGM W OPLB ZIJ **Key 4:** I WILL **Plain Text From Me:** DO HAVE A GEAT DAY (Fig.4)

How I decrypted using vigenere cipher:

I decoded using vigenere cipher tool from dcode.fr, where I converted the cipher text to plain text using the private keys (1, 2, 3, 4) provided by it. After successfully entering the data, I get the received data as shown in figures 1, 2, 3, 4, which are readable text.

Vigenère Cipher Tool for Manual Decryption:

Formula for Vigenère Cipher:

$Pi=(Ci-Ki) \bmod 26$

Where:

Ci = characters of ciphertext (A = 0, B = 1, … Z = 25)

Ki = characters of key (A = 0, B = 1, … Z = 25)

Pi = characters of plaintext

After finding the value of each character, the remainder is divided by 26 to get the plain text.

Cipher Text 1: ECIB AIPBCLU MGF

Key 1: YOUYOUYOUYOU

Plain Text: GOOD MORNING SIR

Step 1: Key iteration matching:

Cipher Text: E C I B A I P B C L U M G F

Key : Y O U Y O U Y O U Y O U Y O

Step 2: Determine the character position (A=0 … Z=25): (Table 1)

A 0	B 1	C 2	D 3	E 4	
F 5	G 6	H 7	I 8	J 9	
K 10	L 11	M 12	N 13	O 14	
P 15	Q 16	R 17	S 18	T 19	
U 20	V 21	W 22	X 23	Y 24	Z 35

Table 1: Values for Alphabets in Vigenere Cipher

Step 1: Determine the letter value (A=0, B=1, … Z=25)

Cipher	**Key**	**C (Cipher value)**	**K (Key value)**	**P = (C-K) mod 26**	**Plain Text**
E	Y	4	24	(4-24+26)=6	G
C	O	2	14	(2-14+26)=14	O
I	U	8	20	(8-20+26)=14	O
B	Y	1	24	(1-24+26)=3	D
A	O	0	14	(0-14+26)=12	M

Cipher	Key	C (Cipher value)	K (Key value)	P = (C-K) mod 26	Plain Text
I	U	8	20	(8-20+26)=14	O
P	Y	15	24	(15-24+26)=17	R
B	O	1	14	(1-14+26)=13	N
C	U	2	20	(2-20+26)=8	I
L	Y	11	24	(11-24+26)=13	N
U	O	20	14	(20-14)=6	G
M	U	12	20	(12-20+26)=18	S
G	Y	6	24	(6-24+26)=8	I
F	O	5	14	(5-14+26)=17	R

Cipher Text 2: WSN FVG RDY UTMPZ

Key 2: PERFECT

Plain Text: HOW ARE YOU DOING

Step 1: Key iteration matching:

Cipher Text: W S N F V G R D Y U T M P Z

Key : P E R F E C T P E R F E C T

Step 2: Character Positioning and Calculation:

Cipher	Key	C (Cipher value)	K (Key value)	P = (C-K) mod 26	Plain Text
W	P	22	15	22-15=7	H
S	E	18	4	18-4=14	O
N	R	13	17	13-17+26=22	W
F	F	5	5	5-5=0	A
V	E	21	4	21-4=17	R
G	C	6	2	6-2=4	E
R	T	17	19	17-19+26=24	Y
D	P	3	15	3-15+26=14	O
Y	E	24	4	24-4=20	U
U	R	20	17	20-17=3	D
T	F	19	5	19-5=14	O
M	E	12	4	12-4=8	I
P	C	15	2	15-2=13	N
Z	T	25	19	25-19=6	G

Step 3: Counting all the characters yields the plain text: **HOW ARE YOU DOING**

Cipher Text 3: OVRGURT QWB SCS IOS

Key 3: OKAY

Plain Text: ALRIGHT SIR SEE YOU

Step 1: Key iteration matching:

Cipher Text: O V R G U R T Q W B S C S I O S

Key : O K A Y O K A Y O K A Y O K A Y

Step 2: Apply the character values (A=0 … Z=25) and the formula:

Cipher	Key	C (Cipher value)	K (Key value)	P = (C-K) mod 26	Plain Text
O	O	14	14	14-14=0	A
V	K	21	10	21-10=11	L
R	A	17	0	17-0=17	R
G	Y	6	24	6-24+26=8	I
U	O	20	14	20-14=6	G
R	K	17	10	17-10=7	H
T	A	19	0	19-0=19	T
Q	Y	16	24	16-24+26=18	S
W	O	22	14	22-14=8	I
B	K	1	10	1-10+26=17	R
S	A	18	0	18-0=18	S

C	Y	2	24	2-24+26=4	E
S	O	18	14	18-14=4	E
I	K	8	10	8-10+26=24	Y
O	A	14	0	14-0=14	O
S	Y	18	24	18-24+26=20	U

Step 2: Plain Text

Cipher Text: O V R G U R T Q W B S C S I O S
Key: O K A Y O K A Y O K A Y O K A Y
Plain Text: **ALRIGHT SIR SEE YOU**

Cipher Text 4: LK PLGM W OPLB ZIJ

Key: I WILL
Plain Text: DO HAVE A GREAT DAY

Cipher Key	C (Cipher value)	K (Key value)	P = (C-K) mod 26	Plain Text Letter	Word
L	I	11	8	11-8=3	D	DO
K	W	10	22	10-22+26=14	O	DO
P	I	15	8	15-8=7	H	HAVE
L	L	11	11	11-11=0	A	HAVE
G	L	6	11	6-11+26=21	V	HAVE

Cipher Key	C (Cipher value)	K (Key value)	P = (C-K) mod 26	Plain Text Letter	Word
M	I	12	8	12-8=4	E	HAVE
W	L	22	11	22-11=11	L	A
O	I	14	8	14-8=6	G	GREAT
P	W	15	22	15-22+26=19	T	GREAT
L	I	11	8	11-8=3	D	GREAT
B	L	1	11	1-11+26=16	Q	GREAT
Z	L	25	11	25-11=14	O	DAY
I	I	8	8	8-8=0	A	DAY
J	W	9	22	9-22+26=13	N	DAY

Final Plain Text

HAVE A GREAT DAY



Fig. 1: Decrypted Message (Cipher Text 1) Using Vigenere Cipher



Fig. 2: Decrypted Message (Cipher Text 2) Using Vigenere Cipher



Fig. 3: Decrypted Message (Cipher Text 3) Using Vigenere Cipher



Fig. 4: Decrypted Message (Cipher Text 4) Using Vigenere Cipher

Task 2 (a):

My teammate gave the text:

MZ UZRADYMFUAZ EKEFQY UE MZ ADSMZULQP EFDGOFGDQ FTMF EGBBADFE PQOUEUAZ YMWUZS NK OAXXQOFUZS BDAOQEEUZS EFADUZS MZP EQZPUZS UZRADYMFUAZE

Where I found that three cipher tools were used for Encryption. The result which I found by performing Cipher Identifier tools, which was shown in figure 5.

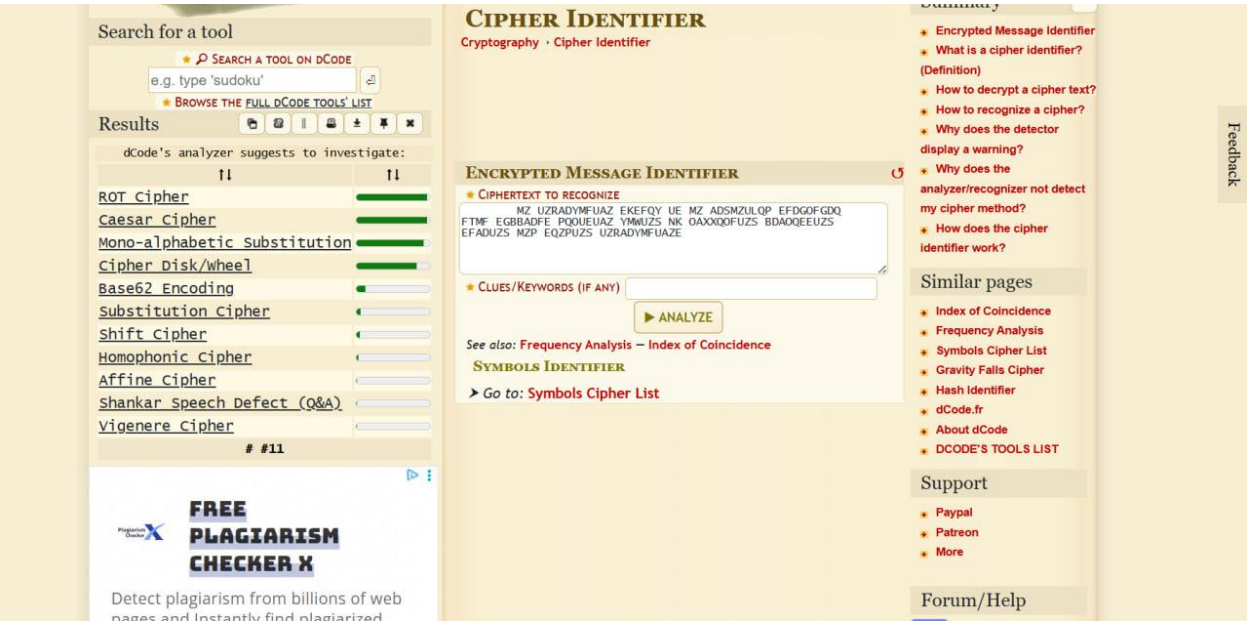


Fig. 5: Cipher Identifier tool was used to identify the decryption tools

1st Perform (ROT Cipher):

From the first perform of using ROT Cipher tool have shown in figure 6, I got the readable text from the first line.

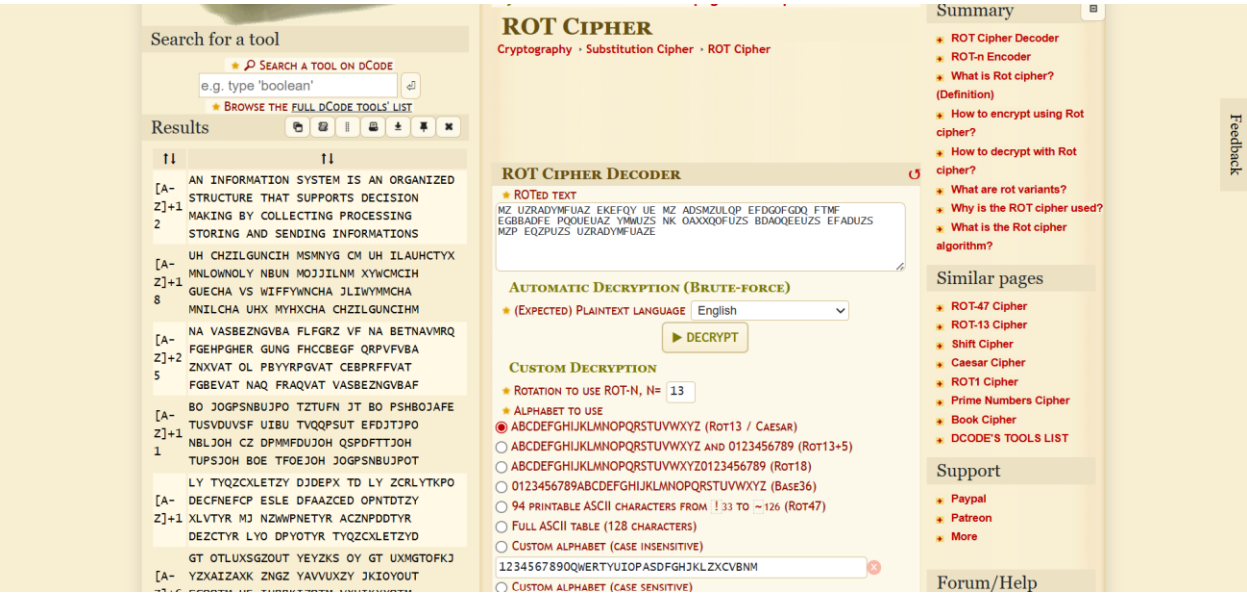


Fig. 6: ROT Cipher tool was for decryption of the message encrypted message

I have found the **first readable text** which signifies the message of my teammate:

AN INFORMATION SYSTEM IS AN ORGANIZED STRUCTURE THAT SUPPORTS DECISION MAKING BY COLLECTING PROCESSING STORING AND SENDING INFORMATIONS

Secondly, I have performed **Caeser Cipher** and after performing the **key from 1 – 12**, I have gotten my message that was sent by **Danny**.

The cipher text was: MZ UZRADYMFUAZ EKEFQY UE MZ ADSMZULQP EFDGOFGDQ FTMF EGBBADFE PQOUEUAZ YMWUZS NK OAXXQOFUZS BDAOQEEUZS EFADUZS MZP EQZPUZS UZRADYMFUAZE

The keys for Caser Cipher are (1-25). After **12** performs or inserting **key 12,** I have found his key 12 and the text after decryption using Caeser Cipher was (Fig. 7): AN INFORMATION SYSTEM IS AN ORGANIZED STRUCTURE THAT SUPPORTS DECISION MAKING BY COLLECTING PROCESSING STORING AND SENDING INFORMATIONS

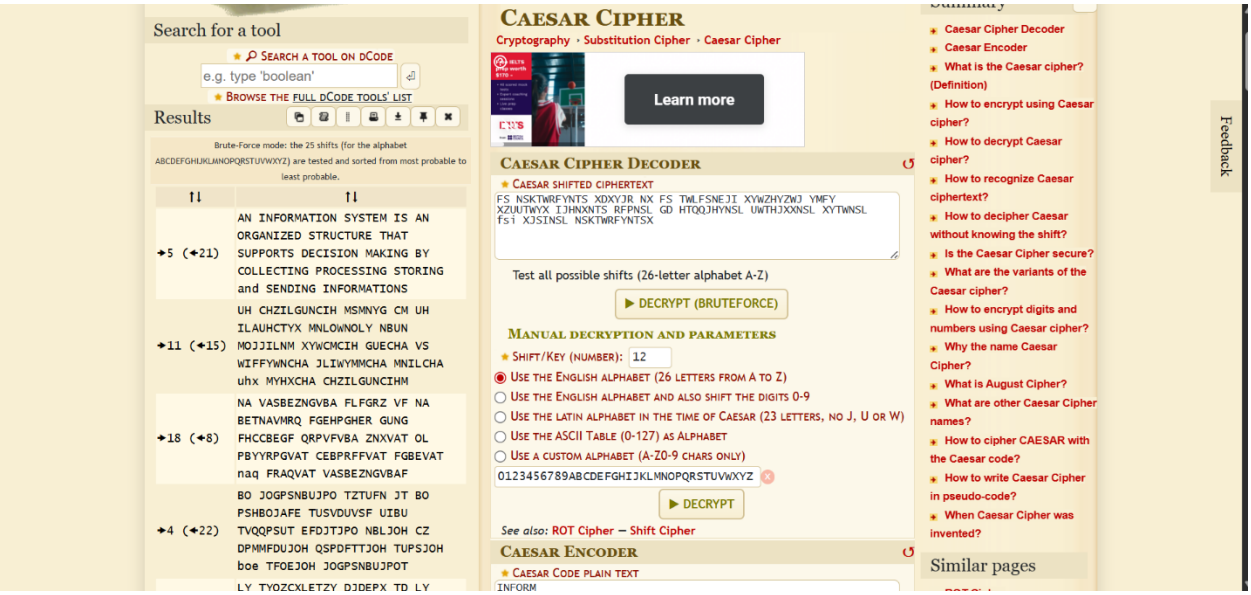


Fig. 7: Caeser Cipher tool was for decryption of the given encrypted message

The message that was given to me from my teammate: AN INFORMATION SYSTEM IS AN ORGANIZED STRUCTURE THAT SUPPORTS DECISION MAKING BY COLLECTING PROCESSING STORING AND SENDING INFORMATIONS

My teammate used **Caeser** and **ROT cipher** tools.

TASK 2 (a)

Plain Text for my teammate for encryption and decryption tools:

I used **ALPHAGRAMS** alphabetic text cipher tool was shown in fig. 8 and I gave him –

Cipher text: T H RE OO M SI UF OLL UTSF TNED WS H RE EE HCA NA VED YRE UTS TNED S TTA NE D NA TODD H E LC WSSA RO SK NA LD WBA RO SK TROF H SE ME TSE RE

Message or Plain text:

My teammate succeed to decrypt my message. The extracted pain text that he gave me was:

THEROOMISFULLOFSTUDENTSWHEREEACHANDEVERYSTUDENTSATTENDANDDOTHECLASSWORKSANDLABWORKSFORTHESEMESTER

My given original message was: THE ROOM IS FULL OF STUDENTS WHERE EACH AND EVERY STUDENTS ATTEND AND DO THE CLASSWORKS AND LABWORKS FOR THE SEMESTER

After adding necessary spaces at text of his decryption, I got my decrypted message to him.

Plain Text	A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
My Text	T H E R O M I S F U L D N W A C V Y K B G J P Q X Z

Steps for **encryption**:

1. I have placed all the alphabet serially in the above table, and also have placed the cipher text's alphabet with the following plain text alphabet orders.
2. For finding the orders for encryption, I have not placed any repeating words once in the table. I have removed duplicates. I have done the mapping of plain text and cipher text.
3. In the last part, I have located the plain text for making encryption (cipher text), ordered according to the plain text's alphabet order and have succeeded in making the encrypted text or cipher text.
4. For checking, I have used ALPHAGRAMS cipher tools at dcode.fr for encryption.

For **decryption:**

I have rebuilt the substitution alphabet and completed mapping cipher text of my given message to my main original plaintext message.

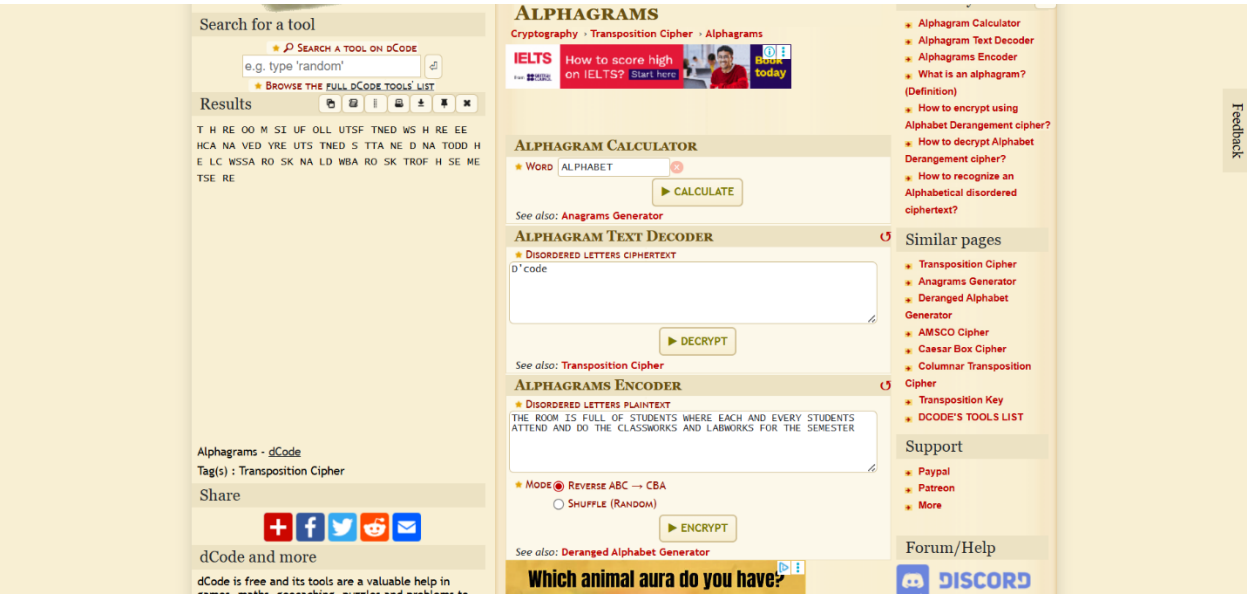


Fig. 8: Shows the encryption of my message to my teammate for decryption.

Alphagrams Cipher Tool is used primarily for speed, accuracy, and educational purposes. Manually calculating the value of each character and decrypting it is time-consuming and prone to errors. Alphagrams tool does these tasks in seconds and allows you to verify the results with manual calculations. It supports multiple types of classical ciphers (such as Vigenère, Caesar, Atbash, Substitution, etc.), so there is no need to use separate tools. It is also useful from an

educational perspective, because each step of cipher breaking is easily explained visually. As a result, it is easy to understand how plain text is derived from cipher text. The biggest advantage is that when calculating by hand, you can cross-check with Alphagrams tool to ensure that the decryption is correct.

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